

Qatar-1b

R-2604

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250927 · created 2026-08-28T00:57:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460945.7438 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.038
Transit depth (Rp/Rs)^2	3.69 +/- 1.45 [%]
Orbital Inclination (inc)	81.7 +/- 1.92
Airmass coefficient 1 (a1)	0.946 +/- 0.015
Airmass coefficient 2 (a2)	0.042 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.59 %
Best Comparison Star	#2 - [357, 73]
Optimal Aperture	3.02
Optimal Annulus	11.7
Transit Duration (day)	0.052 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.038 vs 0.1463 (deviation 1.2σ)
Duration fit vs published	0.052 d vs 0.0692 d (deviation 0.82σ)
Mid-time O–C	-0.6 min
Depth SNR	5.1
Residual scatter	1.59 %

Light curve

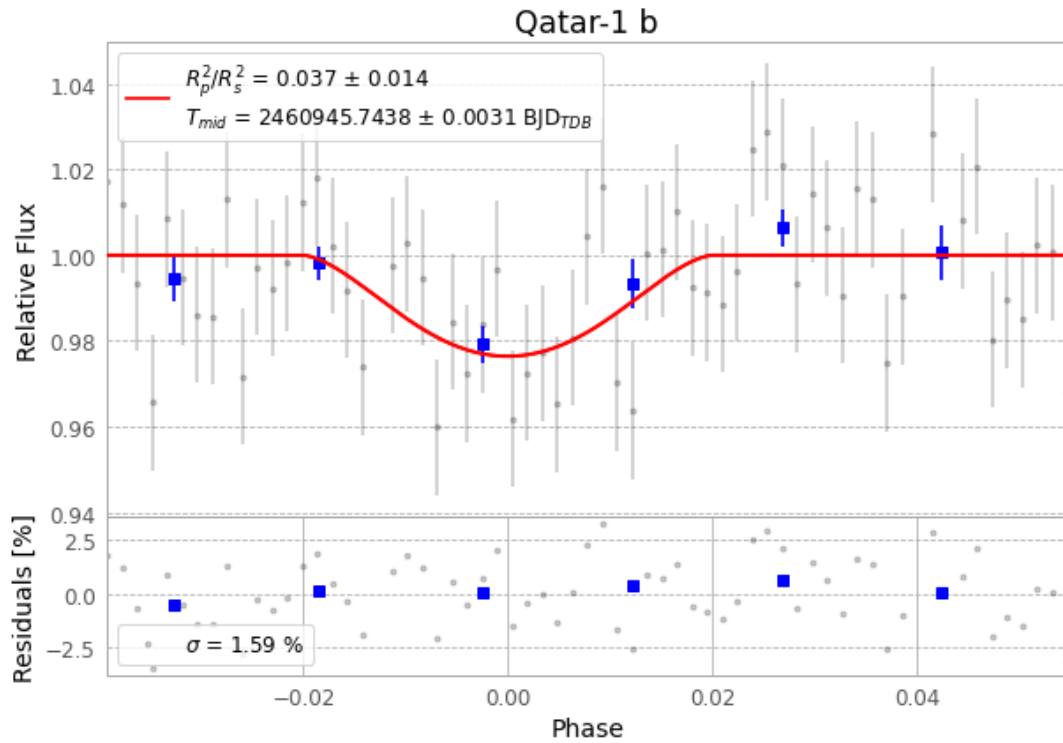


Figure 1 — FinalLightCurve_Qatar-1 b_2025-09-26.png (SHA-256 da814a258d23390c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 334], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f99b383c8c9c5e32...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), Qatar-1250927042715.FITS → Qatar-1250927073916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250927.log (f98c15fad039...), FinalLightCurve_Qatar-1 b_2025-09-26.png (da814a258d23...), FinalLightCurve_Qatar-1 b_2025-09-26.csv (6fb4b38bd562...)

Compute resources

Wall time 123.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:57Z.